Literature Status: Complementability of the Kernels of $AC - I$ and $BA - I$

Run `fa_banach_001` — lane 3

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Status. `literature_already_answered`.

Source question

Let X, Y be Banach spaces and let

$$A \in \mathcal{B}(X, Y), \qquad B, C \in \mathcal{B}(Y, X), \qquad ABA = ACA.$$

Zeng and Zhong, *Common properties of bounded linear operators AC and BA: Spectral theory*, arXiv:1303.5818, state on PDF page 2 that they are "unable to decide whether $AC - I$ and $BA - I$ share common complementability of kernels." Equivalently, is

$$\ker(AC - I_Y) \text{ complemented in } Y \quad \Longleftrightarrow \quad \ker(BA - I_X) \text{ complemented in } X?$$

Decisive later answer

Zeng, Yan, and Zhang, *New results on common properties of the products AC and BA, II*, *Mathematische Nachrichten* **293** (2020), 1629–1635, DOI 10.1002/mana.201900038, explicitly state in item (ii) of the published abstract:

> Under the same hypothesis $ABA = ACA$, $I - AC$ and $I - BA$ share common complementability of kernels and ranges.

The kernel assertion is exactly the source question; replacing $I - T$ by $T - I$ does not change a kernel.

Direct identification check

Set

$$M = \ker(BA - I_X), \qquad N = \ker(AC - I_Y), \qquad D = BAC.$$

The maps $A : X \to Y$ and $D : Y \to X$ restrict to inverse isomorphisms between M and N. Indeed, for $x \in M$ and $y \in N$, the relation $ABA = ACA$ gives

$$AC(Ax) = A(BAx) = Ax, \qquad BA(Dy) = BABA(Cy) = BACA(Cy) = Dy,$$

and

$$DAx = BACAx = BABAx = x, \qquad ADy = ABACy = ACACy = y.$$

Because both inverse maps extend boundedly to the ambient spaces, complementability transfers. If $P : X \to M$ is a bounded projection, then $APD : Y \to N$ is a bounded projection. If $Q : Y \to N$ is a bounded projection, then $DQA : X \to M$ is a bounded projection.

This calculation is included only to verify the exact match between the source question and the literature claim. No novelty is asserted.

Search and source note

The run indexes, local source corpus, exact source wording, titles, DOI, $ABA = ACA$, and close kernel-complementability phrases were searched. The 2020 paper is a separate later publication and explicitly answers the question. The original arXiv PDF is retained as `source_paper.pdf`. The decisive publisher PDF endpoint denied automated access, so the supporting PDF is not copied; the DOI record and published abstract supply the identification.